FREELANCE · SOFTWARE ENGINEERING

Jens Laufer

Production LLM systems — retrieval over real documents, structured outputs and tool calling, evaluations before anything ships

Production LLM systems · RAG over documents · Structured outputs & tool calling · LLM evaluations · TypeScript · Python · Docker · CI/CD · 29 years of software engineering

Since late 2025 I build and operate production LLM systems — not prototypes: agent loops with stop conditions, tool access over MCP, retrieval over internal documents, structured outputs with runtime validation, approval gates in front of anything irreversible, and observability over every run. Two such systems run continuously, one of them running my own company around the clock. Before that ~29 years of software engineering, freelance throughout, much of it on document-heavy business systems, with a second track in machine learning.

AVAILABLE **from 15 September 2026**	WORKS **worldwide**	REMOTE SHARE **95 %** 5 % on-site
HOURLY RATE **125 €/h** excl. VAT	BASED IN **Karlstein am Main** Germany	

FOCUS

Production LLM systems, not prototypes: since 12/2025 two agent systems running continuously — a self-directed AI partner and **Fabrik** (fabrikhq.com), a multi-agent platform for autonomous software development; agent loop, tool use over MCP, approval gates, quality gates, observability, budget and rate-limit control with model routing and fallback.

Document-heavy business systems: invoice management platform for SMEs built and operated over three years (Deutsche Verrechnungstelle, 09/2014–07/2017), including OAuth2 service and debt-collection integration; heating-cost file parsing to an industry standard (Brunata, 22 months).

Taking over existing code under production traffic: payment back-end migrated Java EE → Spring Boot (Condor); Deutsche Telekom configuration platform taken over, extended and operated.

JavaScript front-ends end to end: Vue.js property platform on a Python REST back-end (verkaufe.immobilien, 12 months); React and Vue front-ends alongside Java back-ends — Docker, CI/CD, Playwright end-to-end tests.

Strictly test-driven (TDD, JUnit, Mockito, Playwright/Selenium) across 29 years of software engineering, freelance throughout.

ROLES

Full-stack engineering · Solution design · Backend engineering · DevOps · Build & configuration management

Data analysis · Prototyping · Machine learning · Jenkins plugin development · Maven plugin development

SKILLS

LANGUAGES

Java · JavaScript · TypeScript · Python · R · Groovy

PERSISTENCE / DATABASES

MySQL · MongoDB · PostgreSQL · Oracle · Neo4j · JPA · Hibernate

DEVOPS / CLOUD

Docker · Kubernetes · Rancher · GitLab CI · Jenkins · Maven · Grunt · npm · Artifactory · AWS (EC2, S3, SageMaker, VPC) · Microsoft Azure · DigitalOcean

TESTING

JUnit · TDD · Mockito · Selenium · DBUnit · Playwright

API / INTEGRATION

REST · Swagger · OpenAPI · SOAP · Webservices · CXF · XFire · Kafka · ServiceMix

TOOLING / OTHER

Git · GitLab · Subversion · Jira · Confluence · Kibana · ANTLR · JavaCC · ZXing (QR) · PayPal API · RDF/XML/JSON · SAP (BAPI) · Phonegap · JBoss · Tomcat

MICROSERVICES / BACKEND

Spring Boot · Spring · Spring Data · Spring Integration · Spring Security · OAuth2 · Microservices · Feign · Sleuth · Zipkin · Python Eve · Python Flask · Java EE · EJB

FRONTEND

Vue.js · React · AngularJS · GWT · Quasar · HTML · CSS · Bootstrap

SECURITY / IAM

Keycloak · Spring Security · OAuth2

MACHINE LEARNING

scikit-learn · Keras · DeepLearning4J · CNN · Transfer Learning · Supervised/Unsupervised · k-means · PCA · t-SNE · Reinforcement Learning · OpenAI Gym · Topic Modeling

DATA SCIENCE

pandas · numpy · dplyr · Jupyter · RMarkdown · ggplot · seaborn · Vega-Lite · A/B Testing · Webscraping

PROJECT HISTORY (SELECTION FROM 2014)

since 12/2025
ongoing

Autonomous AI agent systems — Harness & Agentic Engineering

Solytics GmbH · Remote — AI / SaaS

Harness engineering · Agent engineering · AI system architecture · Full-stack engineering · DevOps

Building and running autonomous LLM agent systems as a **harness engineer** — the engineering around the model, not the model itself. The harness is the product: the **agent loop** and its stop conditions, **tool use over MCP**, layered **memory** (durable facts, running context, session state), **approval gates** for anything irreversible, **quality gates** that must pass before an agent may ship, **observability** over every run, **budget and rate-limit control** with model routing and fallback, plus recovery when a run is killed mid-flight. A local multi-model proxy keeps the whole thing independent of any single provider. Two systems in production: a self-directing AI business partner that plans, writes code, tests and deploys around the clock, and **Fabrik** (fabrikhq.com), a multi-agent platform for autonomous software delivery with parallel worker agents. Event-driven orchestration over Telegram, GitHub and systemd; several AI-built SaaS products shipped end to end.

Claude / Anthropic API · LLM-Agenten · Agent Harness · Multi-Agent-Orchestration · Tool Use / MCP · RAG · Prompt Engineering · Python · FastAPI · Vue 3 · Go · Docker · systemd · GitHub Actions · DigitalOcean · SQLite · TDD · TypeScript

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07/2024–06/2025
12 months

Digital property profile for owners

verkaufe.immobilien · Remote — Internet

Full-stack engineering · Solution design · DevOps

Full-stack development of a digital property profile that owners create and maintain themselves: Vue.js front-end, Python back-end (Eve) with a REST API and MongoDB. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines for automated build and deployment. Worked to DevOps principles, owning build and operation of the services in Docker. Test-driven development, automated end-to-end tests with Playwright.

Python · Python Eve · MongoDB · TDD · Vue.js · Docker · GitLab · GitLab CI · REST · Playwright

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07/2022–04/2024
22 months

Front- and back-end, heating cost billing

Brunata · Remote — Industry

Full-stack engineering · Solution design · DevOps

Production server-side services and APIs in Java and Spring Boot: a microservice that parses and generates heating-cost files in the Haweiko-Arge industry format. Designed and built in a microservices architecture with REST interfaces (OpenAPI, Swagger), secured with Spring Security and Keycloak, persisted in MongoDB. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines for automated build and deployment. Worked to DevOps principles, owning build and operation of the services in Docker. Test-driven development (JUnit, Mockito), automated end-to-end tests with Playwright. Front-end development.

Java · Spring Boot · MongoDB · JUnit · TDD · React · Docker · GitLab · GitLab CI · Spring Security · Keycloak · Mockito · OpenAPI · Swagger · REST · Microservices · Playwright

01/2022–06/2022
6 months

Payment back-end migration Java EE → Spring Boot

Condor · Remote — Marketing

Backend engineering

Migrated an EJB-based Java EE payment back-end to Spring Boot and shipped it as a Docker container.

Java · Java EE · JPA · Spring Boot · MySQL · Docker · Mockito · JUnit · TDD

12/2019–12/2020
13 months

Web-based configuration of telecommunications devices

Deutsche Telekom · Darmstadt/Remote — Telecommunications

DevOps · Build & configuration management · Full-stack engineering

Took over an existing prototype for configuring telecommunications devices over a web interface, developed it further and moved it into production. Production server-side services and APIs in Java and Spring Boot in a microservices architecture (REST, Feign, Spring Data/JPA), Vue.js front-end, secured with Keycloak, persisted in MySQL and MongoDB. Worked to DevOps principles, owning build, operation and monitoring of the services: Docker and Kubernetes (Rancher), logging and tracing via Kibana, Sleuth and Zipkin. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines for automated build and deployment. Test-driven development (JUnit, Mockito).

Java · Spring Boot · Docker · Kubernetes · Microservices · REST · Spring Data · JPA · JUnit · TDD · Mockito · Vue.js · MySQL · MongoDB · HTML · CSS · GitLab · GitLab CI · Rancher · Kibana · Feign · Keycloak · Sleuth · Zipkin

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04/2019–08/2019
5 months

Data analysis and modelling at a consumer cooperative

Co-operative Group (Coop) · Manchester — Finance, Retail

Machine learning · Data analysis

Exploratory data analysis, data cleaning, aggregation; unsupervised learning (clustering, feature reduction).

R · Python · Jupyter · RMarkdown · ggplot · scikit-learn · pandas · numpy · dplyr · k-means · PCA · t-SNE · Docker · Git

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08/2017–03/2019
20 months

Connected Van — driver's logbook

Volkswagen · Remote — Automotive

Backend engineering

Microservices for vehicle geolocation and a fuel logbook.

Spring Boot · Docker · Kubernetes · Microservices · REST · Java · Maven · Swagger · OpenAPI · Feign · Sleuth · Zipkin · MySQL · MongoDB · Jira · Confluence · JUnit · TDD · Mockito · Microsoft Azure · Kibana · Spring · Git · Kafka

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07/2015–07/2017
25 months

Invoice management platform (SME)

Deutsche Verrechnungsstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

Platform covering the full invoice life cycle (creation, payment, dunning, debt collection); a range of Spring Boot microservices.

Spring Boot · Docker · Kubernetes · Microservices · REST · Java · Maven · Spring Security · OAuth2 · Spring Integration · MongoDB · Jira · Confluence · JUnit · TDD · Mockito · Selenium · AngularJS · Grunt · npm · Spring · Git

02/2015–06/2015
5 months

Invoice management platform (SME) — debt-collection integration

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

Microservice integrating a debt-collection provider

Spring Boot · Docker · Kubernetes · Microservices · REST · Java · Maven · Spring Security · OAuth2 · Spring Integration · MongoDB · Jira · Confluence · JUnit · TDD · Mockito · Selenium · AngularJS · Grunt · npm · Spring · Git

01/2015–02/2015
2 months

Invoice management platform (SME) — OAuth2 service

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

OAuth2 microservice with an administration front-end.

Spring Boot · Docker · Microservices · REST · Java · Maven · Spring Security · OAuth2 · MongoDB · AngularJS · Git

09/2014–12/2014
4 months

Invoice management platform (SME) — prototype

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering · Solution design · Prototyping · Build & configuration management

Evaluation of microservices and SPA approaches, prototype for an invoice management platform.

Java · JavaScript · HTML · CSS · Spring · Bootstrap · AngularJS · Grunt · npm · Spring Boot · Docker · Kubernetes · Microservices · REST · Maven · Spring Security · MongoDB · Jira · Confluence · Git

01/2014–08/2014
8 months

Vehicle configurator prototype

Audi · Remote — Automotive

Full-stack engineering · Solution design · Prototyping · Build & configuration management

SPA on AngularJS (front-end) with a Spring back-end and a RESTful API; comparison of persistence approaches (MySQL, MongoDB, Neo4j).

Java · JavaScript · HTML · CSS · Bootstrap · Spring · Spring Security · Spring Data · MongoDB · Neo4j · MySQL · Maven · Tomcat · JUnit · TDD · Mockito · Selenium · Jira · Git

CONTACT & CREDENTIALS

CONTACT & PROFILES

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GITHUB	github.com/jenslaufer
LINKEDIN	linkedin.com/in/jenslaufer
KAGGLE	kaggle.com/jenslaufer
PUBLICATION	Towards Data Science — author

EDUCATION

11/1997
Dipl.-Ing. (Univ.) Electrical Engineering (MSc equivalent)
University of Erlangen-Nuremberg

05/1989
Abitur (university entrance qualification)
Gymnasium am Romäusring, Villingen-Schwenningen

CERTIFICATES

12/2018

Machine Learning Engineer Nanodegree

Udacity

Verify →

08/2017

Data Analyst Nanodegree

Udacity

Verify →